2023

CY-3001 Network and Cybersecurity-II

Final Project Proposal

Arsh Malik (21i-2752)
Mahareeb Fatima(21i-1564)
Maheen Mansha(21i-1920)

FAST NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Network Security Analysis with Splunk

Introduction

Network-based threats continue to pose serious risks to organizations. This project aims to enhance network security by leveraging Splunk, a powerful data analytics platform, to analyze network traffic logs generated by intrusion detection systems (IDS) which in our case is ***Snort***. This approach will improve the efficiency of cyber security operations and enable faster responses to potential threats.

Objectives

The primary objectives of our project that we have identified for now are as follows:

1. <u>Downloading and Installing Splunk Enterprise:</u>

To start off, we would be installing Splunk, which is a tool that allows its users to monitor, search and analyze machine generated data and logs through a web interface. For basic configuration, we would be configuring the receiving of data that would enable Splunk to receive data forwarded by other instances. The other important thing we would be doing is to configure the Splunk Universal Forwarder. The reason for using SUF is because it would automatically forward the latest logs. Even while the IDS (snort) is running, it forwards those alerts and logs to splunk, and we would be able to see them in real time. The last configuration we would be doing on splunk would be installing the snort app for splunk.

2. <u>Creating and Configuring Splunk Dashboard:</u>

Although the snort app for splunk already comes pre-configured with dashboard, we would still be developing a custom dashboard in Splunk to visualize data and logs generated by Snort that would be forwarded into it through ***Splunk Universal Forwarder***. We will use ***Splunk's Search Processing Language (SPL)*** to build visualizations such as charts, graphs, and tables that provide insights into network activities to make the visualization of the machine generated data more readable and appealing.

3. Configuring Instances:

Next up, we would be configuring two instances that we would be utilizing further in the project. One instance would be the machine where both our snort and webserver would be configured. The other would be the attackers' machine which would be used to employ network-based attacks and traffic on our main machine. This would be for the generation of log files and alerts.

4. Installing and Configuring Snort:

As mentioned above, we would be using snort on our main machine. For that we would install and configure Snort to ensure that it is configured to generate logs containing information about network traffic and security events. For that we would implement rules and signatures within snort to detect and generate alerts for specific network-based threats and activities.

5. Integrating Splunk with IDS (Snort):

Once all the machines are configured separately, we would establish seamless integration between Splunk and Snort. We would configure snort to forward logs and alerts to Splunk for analysis which has already been explained above.

6. Parse and Index Data in Splunk:

Additionally, we will configure Splunk to parse and index incoming data from snort to make it searchable and actionable. For that we will define source types, timestamps, and field extractions to ensure accurate and meaningful representation of the data within Splunk.

7. Documentation and Training:

The last thing we would be doing is documentation of the entire process in detail. This would be helpful for any future reference.

Expected Outcomes

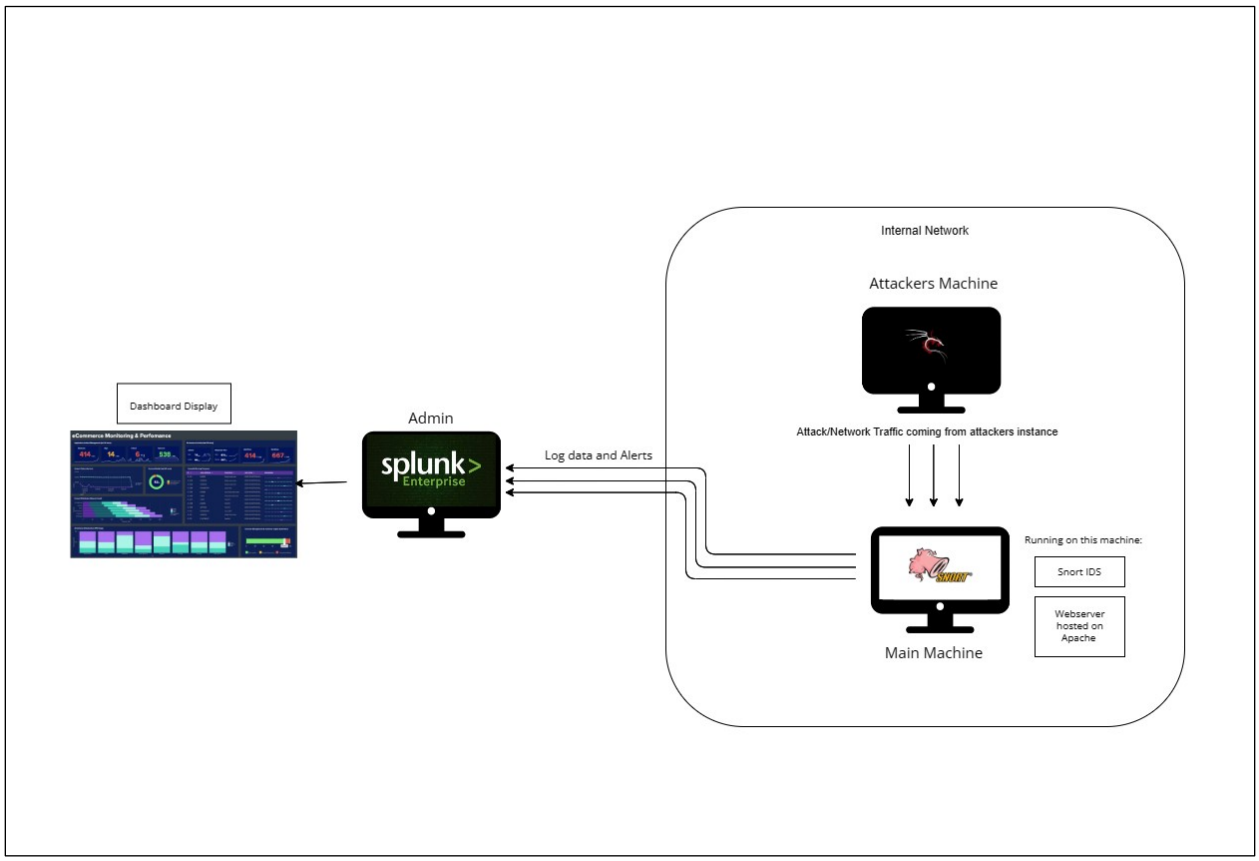
Upon successful completion of the project, we have the following expected outcomes:

- A well-designed Splunk dashboard displaying insights into network security events.
- Seamless integration between Splunk and the chosen IDS for efficient data flow.
- Generation of meaningful logs and alerts for network-based threats and activities.
- Centralized storage and analysis of network security data in Splunk.
- Improved efficiency and responsiveness of cyber security operations.

Tools to be Used

- Splunk Enterprise
- Snort (IDS)
- Any cloud hosting service to host splunk like AWS, Linode etc. if possible

Visual Representation



Conclusion

Automating network security analysis with Splunk represents a significant step towards fortifying organizations against evolving cyber threats. By integrating Splunk with any IDS, we enhance the ability to detect, analyze, and respond to network-based threats promptly, ultimately fostering a more secure online environment for both individuals and enterprises.